CustomerSegment ML

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

PROJECT REPORT

Submitted by

HERSHLY BOSE J [RA2411026040096]

DEENA DARSHAN M [RA2411026040097]

LIJO Y [RA2411026040074]

Under the guidance of

S.POORNIMA

(Assistant Professor (JrG), Department of CSE - Emerging Technologies)

in partial fulfillment for the Course

of

21CSC305P - MACHINE LEARNING

in

COMPUTER SCIENCE AND ENGINEERING

WITH A SPECIALIZATION IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(EMERGING TECHNOLOGIES)

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

VADAPALANI CAMPUS

NOVEMBER 2026

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

Certified that this project report titled "**CustomerSegment ML: Customer Segmentation Using K-Means Clustering**" is the Bonafide work of **"HERSHLY BOSE J [RA2411026040096]", "DEENA DARSHAN M [RA2411026040097]", "LIJO Y [RA2411026040074]"** who carried out a project work in 21CSC305P - Machine Learning under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE	**SIGNATURE**
Ms. S.Poornima, M.Tech, (Ph.D),	**Dr. P. Chitra, ME, Ph.D,**
Assistant Professor(Jr.G)	Professor & HOD
Dept. of CSE	Dept. of CSE
(Emerging Technologies)	(Emerging Technologies)

ACKNOWLEDGEMENTS

We express our humble gratitude to our Honorable Chancellor **Dr. T. R. Paarivendhar,** Pro Chancellor (Administration), **Dr. Ravi Pachamuthu,** Pro Chancellor (Academics) **Dr. P. Sathyanarayanan,** Pro Chancellor (Admin) for the facilities extended for the completion of the minor project work.

We would record our sincere gratitude to our Vice Chancellor, **Dr. C. Muthamizhchelvan** and Registrar, **Dr. S. Ponnusamy,** for their support in completing our minor project work by giving us the best of academic excellence support system in place. We extend our sincere thanks to our Director Academics, **Dr.Rama S Komaragiri** & Director Administration **Dr. Venkatesh Babu** of SRM Group. We extend our sincere thanks to our Dean Faculty of Engineering and Technology - **Dr. C.V. Jayakumar,** and Vice Principal - Academics **Dr. C. Gomathy** and Vice Principal - Examination - **Dr. S. Karthikeyan** for their invaluable support.

We wish to thank **Dr. P. Chitra**, Professor & Head, Department of Computer Science & Engineering - Emerging Technologies, SRM Institute of Science and Technology, Vadapalani Campus for her valuable suggestions and encouragement throughout the period of the minor project work and the course.

We extend our gratitude to our course faculty, **S.Poornima** Assistant Professor (Jr.G), Department of Computer Science & Engineering (Emerging Technology), SRM Institute of Science and Technology, Vadapalani Campus for providing us an opportunity to pursue our project under her mentorship. She provided us with the freedom and support to explore the research topics of our interest. We sincerely thank all the faculty and students of the department who have directly or indirectly helped our minor project. Finally, we would like to thank our parents, our family members and our friends for their unconditional love, constant support, and encouragement.

TEAM MEMBERS:

HERSHLY BOSE J [RA2411026040096]

DEENA DARSHAN M [RA2411026040097]

LIJO Y [RA2411026040074]

ABSTRACT

A small gift shop sees very different customers every day. A college student may come back every week for a new key chain, a working adult may buy a bouquet once a month, and an elderly customer may visit only for a wedding or a temple festival. Sending the same offer to all of them wastes money and misses the customers most likely to return. **Customer segmentation** solves this by dividing customers into groups whose members behave alike. This project, **CustomerSegment ML**, is a complete machine-learning system that performs this segmentation automatically on any customer file that a shop owner uploads.

The system is built around **K-Means clustering**, an unsupervised algorithm that needs no labelled data. The case study is a Tamil Nadu gift shop that sells **Key Chains, Bouquets and Miniature Toys**. A customer CSV with Name, Age, Gender, District, Product_Type and Repeat_Purchases is uploaded through a **Next.js** dashboard and processed by a **FastAPI** service. The service validates the file, standardises the two behavioural features, **Age and Repeat Purchases**, sweeps K from 2 to 8, and recommends the number of clusters with the highest **silhouette score**. K-Means is fitted with k-means++ seeding and ten restarts, and every cluster is described by a persona, its top product, its top district and a marketing strategy. New customers are classified in real time.

Three synthetic datasets of **70, 85 and 100 customers** drawn from 20 Tamil Nadu districts were generated with hidden ground-truth groups. On all three the system recommends the correct number of groups (3, 4 and 5) and reaches **100%, 98.8% and 99% accuracy**. On the 100-customer file it obtains a silhouette score of **0.6654**, a Davies-Bouldin index of **0.442** and a Calinski-Harabasz index of **331.4**. Compared with thirteen alternative and baseline methods it ranks **first on silhouette, Calinski-Harabasz and accuracy**, far ahead of grouping by product type (33%) and random assignment (30%). It produced the same partition on all 30 random seeds tested.

Keywords: customer segmentation, K-Means clustering, unsupervised learning, repeat purchases, silhouette analysis, elbow method, retail analytics, FastAPI, Next.js.

TABLE OF CONTENTS

S. No.	TITLE	PAGE NO.
1	**CHAPTER 1: INTRODUCTION**	**6**
	1.1 Background of the study	6
	1.2 Problem Statement	6
	1.3 Objectives	7
	1.4 Scope and Significance	7
2	**CHAPTER 2: LITERATURE REVIEW**	**8**
	2.1 Related Work	8
	2.2 Comparison of Existing Techniques	9
3	**CHAPTER 3: METHODOLOGY**	**11**
	3.1 Data Collection	11
	3.2 Data Preprocessing	11
	3.3 Feature Selection / Extraction	12
	3.4 Model Selection	13
	3.5 Algorithm Used	14
	3.6 Model Training & Validation	15
4	**CHAPTER 4: SYSTEM DESIGN**	**17**
	4.1 Architecture Diagram	17
	4.2 Flow Chart	18
	4.3 Module Description	19
5	**CHAPTER 5: IMPLEMENTATION**	**20**
	5.1 Tools and Technologies Used	20
	5.2 Model Development Steps	20
6	**CHAPTER 6: RESULTS & DISCUSSION**	**22**
	6.1 Experimental Results	22
	6.2 Performance Comparison	28
	6.3 Result Analysis	30
7	**CHAPTER 7: CONCLUSION & FUTURE WORK**	**32**
	7.1 Summary of Findings	32
	7.2 Limitations	32
	7.3 Future Enhancements	32
8	**REFERENCES**	**33**
9	**APPENDICES**	**35**
	9.1 Dataset Description	35
	9.2 Code Snippets	35
	9.3 Additional Graphs & Outputs	37

CHAPTER 1: INTRODUCTION

1.1 Background of the study

Small and medium retailers in Tamil Nadu, from gift shops in Chennai to stalls near temples in Madurai and Thanjavur, depend on customers who come back. A customer who returns ten times is worth far more than one who buys once, yet most small shops treat every customer the same way. The classic answer from marketing theory, formalised by Smith in 1956 [14], is **market segmentation**: divide the customers into a small number of groups that behave alike and design one plan per group.

Shop owners usually segment by instinct or by a single rule, such as “students” versus “elders”. Such rules look at one variable at a time and miss how behaviour actually combines. A young customer can be a loyal weekly visitor or someone who bought once for a friend's birthday. **Machine learning** offers a better route. Clustering algorithms examine several variables together and discover the natural groups hidden in the data without anyone defining them in advance [13], [15].

Among clustering methods, **K-Means** [1], [2] is the most widely used in industry. It is easy to explain, fast, and produces a *centroid* for each group that reads directly as the profile of a typical member, for example “about 22 years old, 15 repeat purchases”. Repeat purchases are a direct measure of loyalty and are closely related to the *frequency* dimension of the well-known RFM model [16].

This project, **CustomerSegment ML**, turns that idea into a usable product for a gift shop selling **Key Chains, Bouquets and Miniature Toys**. A shop owner uploads a customer file and receives an interactive dashboard with the elbow curve, a clustered scatter plot of age against repeat purchases, a card for every group with its top product, top district and a marketing strategy, a live prediction form for new customers and a downloadable file of results. It was developed as the course project for 21CSC305P Machine Learning.

1.2 Problem Statement

Given a table of customers in which each row contains at least an **age** and a **number of repeat purchases** (with optional name, gender, district and product type), partition the customers into K groups such that customers inside a group are as similar as possible and customers in different groups are as different as possible, choose K automatically, describe each group in business terms, and assign any future customer to the correct group instantly. Formally this is an **unsupervised clustering** problem: there are no labels, so the quality of a solution must be judged by internal indices of cohesion and separation, and, for our synthetic data, by comparison with the hidden groups used to generate it.

Several practical difficulties make the problem more than a textbook exercise:

- **No labels.** Real shops do not know the “correct” groups, so the system must choose the number of clusters itself and justify that choice.

- **Different feature scales.** Age ranges from about 16 to 62 years while repeat purchases range from about 1 to 19. Without scaling, age would dominate the distance calculation.
- **Small, messy files.** A small shop may have only 70 to 100 customers, and files can contain blank, negative or non-numeric values that must be skipped without crashing.
- **Sensitivity to initialisation.** K-Means can converge to a poor solution if its starting centroids are badly placed.
- **Interpretability.** A cluster number means nothing to a shop owner; each group must be explained and paired with an action.

1.3 Objectives

- Build an end-to-end pipeline that turns an uploaded customer CSV into described customer segments with no manual steps.
- Detect the Age and Repeat_Purchases columns automatically, keep Name, Gender, District and Product_Type for description, and skip invalid rows with a clear report.
- Compute the elbow curve (WCSS) for K = 2 to 8 and recommend the K with the highest silhouette score.
- Train K-Means with k-means++ seeding and ten restarts to obtain a stable, reproducible partition.
- Describe every cluster with a persona, its most common product and district, and a marketing strategy suited to its age group and loyalty.
- Provide real-time prediction of a new customer's group from age and repeat purchases.
- Generate realistic Tamil Nadu gift-shop datasets of 70 to 100 customers and evaluate the model with silhouette, Davies-Bouldin, Calinski-Harabasz and accuracy against the hidden groups, compared with alternative and baseline methods.

1.4 Scope and Significance

The scope of the project is a decision-support tool for small retailers that clusters customers on two behavioural features, age and repeat purchases, for files of 10 to 5,000 customers. It contributes a reusable, tested segmentation engine, automatic model selection, human-readable personas with product and district insights, a validated REST API and a modern dashboard with charts, tables, a K slider, a prediction form and CSV export.

The significance of the work lies in closing the gap between a machine-learning algorithm and a shop owner's decision: segmentation that can be run on the shop's own file in seconds, inspected visually and downloaded is far more likely to change how the shop spends its marketing effort than a notebook, and the same pipeline is a clear teaching example of unsupervised learning.

CHAPTER 2: LITERATURE REVIEW

Clustering is one of the oldest problems in statistics and data mining, and customer segmentation is one of its most common commercial uses. This chapter reviews the clustering algorithms relevant to the project, the methods used to choose the number of clusters and validate a partition, and the use of loyalty and purchase behaviour in segmentation. It ends with a structured comparison that positions CustomerSegment ML against the existing techniques.

2.1 Related Work

Partitional clustering: K-Means and its variants

K-Means was described by MacQueen in 1967 [1], while the iterative assign-and-update procedure that most libraries implement is known as **Lloyd's algorithm** [2]. It minimises the within-cluster sum of squared distances and usually converges in a handful of iterations. Its main weakness is sensitivity to the starting centroids. Arthur and Vassilvitskii addressed this with **k-means++** [3], which spreads the initial centroids by choosing each new seed with probability proportional to its squared distance from the seeds already chosen, giving an expected solution within O(log K) of the optimum. **Mini-Batch K-Means** [12] updates centroids from small random batches and is designed for web-scale data.

Hierarchical clustering

Agglomerative methods start with every point in its own cluster and repeatedly merge the two closest clusters. **Ward's method** [8] merges the pair that causes the smallest increase in total within-cluster variance and therefore behaves much like K-Means. Complete and average linkage use the farthest and the average pairwise distances, while single linkage uses the nearest pair and tends to produce long chained clusters. Hierarchical methods give a full dendrogram but cannot assign a new point without re-running the merge.

Density-based and model-based clustering

DBSCAN [9] groups points that lie in dense regions and labels isolated points as noise; it finds clusters of arbitrary shape but depends strongly on its radius parameter. **Gaussian Mixture Models** fitted with the EM algorithm [10] treat each cluster as a Gaussian distribution and give soft memberships, at the cost of more parameters. **BIRCH** [11] builds a compact clustering-feature tree for very large databases.

Choosing K and validating clusters

Because clustering has no labels, internal indices judge quality. The **elbow method**, dating back to Thorndike [7], plots the within-cluster sum of squares against K and looks for the point where adding clusters stops paying off. The **silhouette coefficient** of Rousseeuw [4] compares each point's average distance to its own cluster with its distance to the nearest other cluster. The **Davies-Bouldin index** [5] (lower is better) and the **Calinski-Harabasz index** [6] (higher is better) measure the balance between within-cluster scatter and between-cluster separation. The **gap statistic** [23] compares WCSS with uniform random data, and the **adjusted Rand index** [24] measures agreement between two partitions.

Loyalty and purchase behaviour in segmentation

Marketing research has long used demographic and behavioural variables to build segments [15]. **RFM analysis** [16] scores customers on recency, frequency and monetary value, and its frequency component, how often a customer buys again, is one of the strongest predictors of future value. Combining age with repeat-purchase frequency is therefore a natural way to separate loyal fans from occasional visitors in different age groups. Most published work, however, applies clustering in a notebook to large supermarket or online datasets; few studies package it as a validated tool that a small shop can run on 70 to 100 customers.

2.2 Comparison of Existing Techniques

Table I compares representative techniques with the proposed CustomerSegment ML system.

TABLE I
COMPARISON OF EXISTING TECHNIQUES WITH THE PROPOSED WORK

S. No.	Technique	Methodology	Advantages	Limitations Compared with the Proposed Work
1	Rule-based bands / RFM [16]	Fixed thresholds on one or a few variables	Very simple to explain	Ignores joint structure; thresholds are arbitrary
2	K-Means, random init [1], [2]	Lloyd iterations from random seeds	Fast, centroid profiles	Unstable; can fall into poor local minima
3	k-means++ [3]	D^2 weighted seeding before Lloyd	Better, provable seeding	Still needs K chosen by the user
4	Mini-Batch K-Means [12]	Centroid updates from random batches	Scales to huge data	No benefit on small shop files
5	Agglomerative (Ward / linkage) [8]	Bottom-up merging of clusters	Dendrogram, no seeding	Cannot predict new customers directly
6	DBSCAN [9]	Density reachability with eps radius	Arbitrary shapes, finds noise	Sensitive to eps; fragments small data
7	Gaussian Mixture (EM) [10]	Probabilistic mixture of Gaussians	Soft memberships	More parameters; harder to explain
8	BIRCH [11]	CF-tree summarisation then clustering	Memory efficient for big data	Threshold tuning; weaker on small sets
9	**Proposed CustomerSegment ML**	**Scaled K-Means on age and repeat purchases, k-means++, 10 restarts, silhouette-selected K, personas, API + dashboard**	**Automatic K, stable, explainable, product and district insights, real-time prediction**	**Uses two behavioural features in this version**

Summary of the comparison

No single algorithm is best in every situation, but for compact groups of customers K-Means with good seeding is accurate, fast and uniquely easy to explain through its centroids. Its weaknesses, choosing K and sensitivity to initialisation, are handled by silhouette-based selection and by k-

means++ with several restarts. CustomerSegment ML combines these remedies with data validation, product and district summaries and a real-time service, and Chapter 6 measures it against the alternatives instead of assuming it is better.

CHAPTER 3: METHODOLOGY

3.1 Data Collection

Real customer records of small shops are private and rarely published, so the team generated **synthetic but realistic datasets** for a gift shop in Tamil Nadu. A generator script (`dataset/generate_datasets.py`) draws every customer from one of five behavioural groups, each with a typical age, a typical number of repeat purchases and a preferred mix of products. Names are common Tamil names, districts are drawn from 20 Tamil Nadu districts and products are Key Chains, Bouquet and Miniature Toys. Fixed random seeds make the files reproducible.

Three files were produced: `tn_gift_shop_70.csv` (70 customers, 3 hidden groups), `tn_gift_shop_85.csv` (85 customers, 4 groups) and `tn_gift_shop_100.csv` (100 customers, 5 groups). The files contain only six columns; the hidden group of each customer is **not** written to the files and is kept by the generator only to measure accuracy after clustering. Table II summarises the 100-customer file, which is used as the main evaluation set.

TABLE II
SUMMARY OF THE EVALUATION DATASET (TN_GIFT_SHOP_100.CSV)

S. No.	Attribute	Type	Range / Values	Used for clustering
1	Name	Text	Tamil first name and initial	No
2	Age	Integer	16 to 62 years (mean 38.2)	Yes
3	Gender	Categorical	Female 51, Male 49	No
4	District	Categorical	20 Tamil Nadu districts	No (profiles)
5	Product_Type	Categorical	Bouquet 42, Miniature Toys 34, Key Chains 24	No (profiles)
6	Repeat_Purchases	Integer	1 to 19 (mean 9.0)	Yes

The five hidden groups are **Loyal Young Fans** (about 21 years, 15 repeats, mostly key chains and toys), **Occasional Young Buyers** (about 23 years, 3 repeats), **Regular Adult Customers** (about 37 years, 9 repeats), **Occasional Senior Buyers** (about 56 years, 3 repeats, mostly bouquets) and **Loyal Senior Gifters** (about 55 years, 15 repeats). Because clustering is unsupervised there is no train/test split of labels: the whole file is clustered and quality is measured with internal indices and with accuracy against the hidden groups.

3.2 Data Preprocessing

Uploaded files are often inconsistent, so preprocessing is tolerant of variation while being strict about what the model needs:

1. **Size and encoding check.** The request body must be non-empty and at most 5 MB, decoded as UTF-8 with an optional byte-order mark.

2. **Column detection.** Each header is lower-cased and stripped of non-alphanumeric characters, so `Repeat_Purchases`, `repeat purchases` and `Repeats` all match the repeat field; the same rule matches name, age, gender, district (or city) and product type.
3. **Type coercion and cleaning.** Age and repeat purchases are converted to numbers; blank, non-numeric, infinite or negative values make the row invalid and it is dropped and counted.
4. **Size limits.** At least 10 valid customers are required so that K up to 8 is meaningful, and at most 5,000 are accepted.
5. **Feature scaling.** Age and repeat purchases are standardised with z-score normalisation.

K-Means is entirely based on Euclidean distance, so scaling matters. Each feature x is transformed with its mean μ and standard deviation σ:

$$z = (x - \mu) / \sigma \tag{1}$$

After scaling both features have mean 0 and standard deviation 1. In the 100-customer file age has a standard deviation of 15.1 years while repeat purchases have 5.6, so without scaling a 3-year age difference would count as much as 3 extra visits. Fig. 1 shows the data before and after scaling.

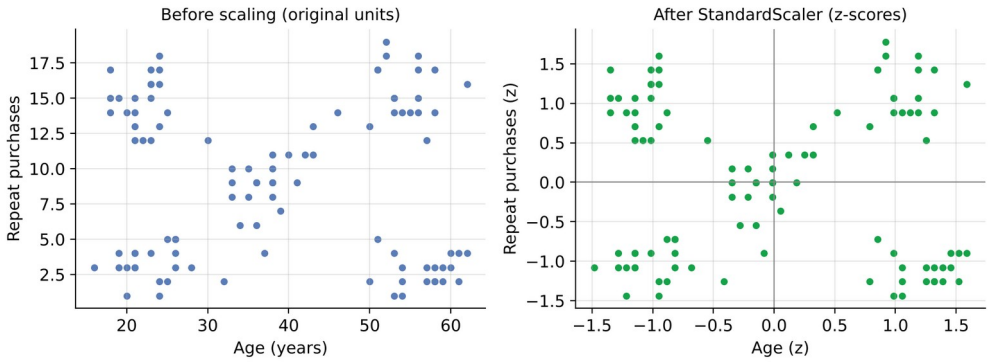


Fig. 1. Customer data before and after z-score standardisation

3.3 Feature Selection / Extraction

Fig. 2 shows the distribution of age, repeat purchases and product type. Age has clusters of young and older customers with fewer people in between, repeat purchases are spread from 1 to 19, and Bouquet is the most common product.

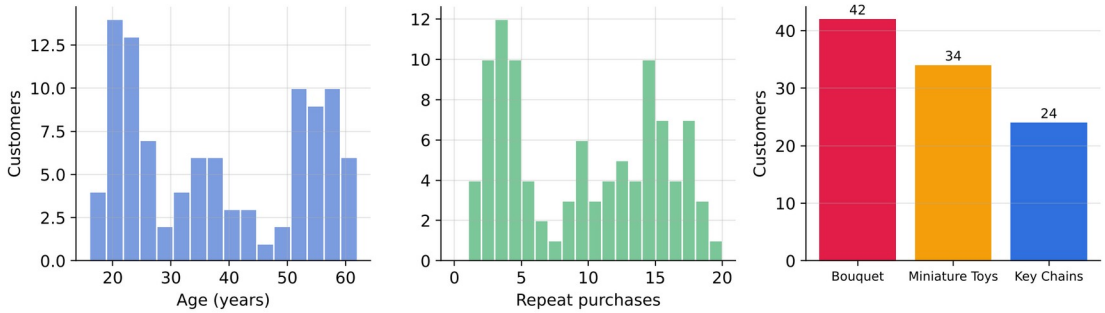


Fig. 2. Distributions of age, repeat purchases and product type

Fig. 3 plots every customer by age and repeat purchases, coloured by product. Five groups are already visible: two young groups (one loyal, one occasional), one adult group in the middle and two senior groups. Age and repeat purchases are almost uncorrelated (r = -0.042), so each carries independent information: who the customer is and how loyal they are.

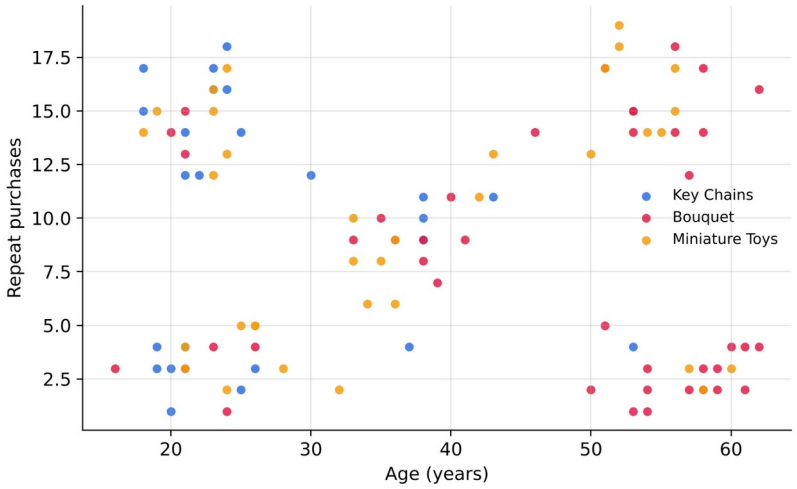


Fig. 3. Customers by age and repeat purchases, coloured by product type

The project therefore clusters on **Age** and **Repeat_Purchases**. Product type, district and gender are kept as **descriptive attributes**: they are summarised for every cluster (top product, top district, share of female customers) but do not drive the grouping. Product type alone is a poor basis for segmentation, as Chapter 6 shows, and gender is a sensitive attribute that should not drive marketing decisions on its own. Two features also mean every result can be drawn exactly on a two-dimensional chart.

3.4 Model Selection

Choice of algorithm

K-Means was selected as the primary model because the groups in Fig. 3 are compact and roughly round, which is exactly the shape K-Means is designed for; its centroids are directly interpretable as the typical age and loyalty of a group; a fitted model classifies a new customer instantly by finding the nearest centroid; and it is fast and deterministic once the seed is fixed. All alternatives in Table I are evaluated on the same data in Section 6.2.

Choice of K

K is chosen automatically by a sweep over K = 2 to 8 (Fig. 4). For every K the system fits K-Means, records the WCSS for the elbow chart and computes the silhouette score. The **recommended K is the one with the highest silhouette score**, because silhouette rewards tight clusters and wide gaps, while WCSS always decreases as K grows.



Fig. 4. Automatic selection of the number of clusters

$$K^* = \arg\max S(K), \quad K \in \{2, 3, \ldots, 8\} \tag{2}$$

Davies-Bouldin and Calinski-Harabasz indices are computed as supporting evidence, and the shop owner can still override the choice with a slider.

3.5 Algorithm Used

The core algorithm is K-Means with k-means++ initialisation, as implemented by scikit-learn [17]. Given n scaled customers $x_1 \ldots x_n$, each a vector [age, repeat purchases], and a number of clusters K, K-Means looks for centroids $\mu_1 \ldots \mu_K$. The distance between a customer and a centroid is the Euclidean distance:

$$d(x, \mu) = \sqrt{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2} \tag{3}$$

Assignment step. Every customer is assigned to its nearest centroid:

$$c_i = \arg\min_j \| x_i - \mu_j \|^2 \tag{4}$$

Update step. Every centroid moves to the mean of the customers assigned to it:

$$\mu_j = (1 / |C_j|) \cdot \Sigma\, x_i , \quad x_i \in C_j \tag{5}$$

Both steps lower the **within-cluster sum of squares (WCSS)**, so the procedure always converges:

$$J = \Sigma_{j=1}^{K} \Sigma_{i \in C_j} \| x_i - \mu_j \|^2 \tag{6}$$

k-means++ seeding. The first centroid is a random customer; each further centroid is chosen with probability proportional to the squared distance $D(x)^2$ from the nearest centroid already chosen:

$$P(x) = D(x)^2 / \Sigma_{x'} D(x')^2 \tag{7}$$

The whole procedure runs **ten times** (`n_init=10`) and the run with the lowest WCSS is kept, with `random_state=42` for reproducibility. Fig. 5 summarises the loop. Centroids are mapped back to real units with $\hat{x} = \sigma \cdot z + \mu$, so they read as “37.1 years, 9.3 repeats”.

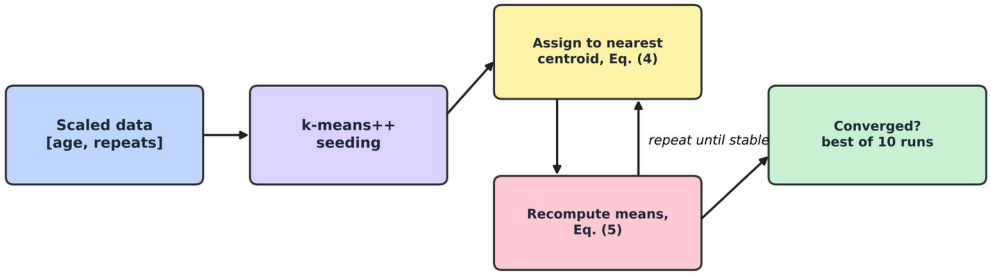


Fig. 5. K-Means with k-means++ seeding and multiple restarts

Persona rule. Each cluster's average age is compared with the overall mean: more than half a standard deviation below is *young*, more than half above is *senior*, otherwise *adult*. Repeat purchases are banded the same way into *occasional*, *regular* and *loyal*. The nine combinations give names such as Loyal Young Fans or Occasional Senior Buyers, each with a matching strategy; for example young groups get Instagram and WhatsApp offers and senior groups get phone, SMS and festival-season offers. The complete algorithm is:

1. Read and validate the uploaded CSV; detect columns and drop invalid rows.
2. Extract X = [age, repeat purchases] and standardise it.
3. For K = 2 … 8: fit K-Means, store WCSS and silhouette; recommend the K with the highest silhouette.
4. Fit the final K-Means (k-means++, 10 restarts, seed 42).
5. Inverse-transform the centroids; compute count, share, mean age, mean repeats, top product, top district and female share per cluster.
6. Assign persona and strategy from the age and loyalty bands; number duplicate names.
7. If the file has a True_Segment column, compute accuracy, ARI and NMI against it.
8. For a new customer: scale [age, repeats], find the nearest centroid and return its persona and strategy.

3.6 Model Training & Validation

Validation uses internal indices and, for the synthetic files, external agreement with the hidden groups. For a customer i, let a(i) be its mean distance to the other members of its own cluster and b(i) its mean distance to the nearest other cluster. The silhouette coefficient [4] and its mean are:

$$s(i) = (b(i) - a(i)) / \max\{ a(i), b(i) \} \quad (8)$$

$$S = (1 / n) \Sigma_{i=1}^{n} s(i) \quad (9)$$

The **Davies-Bouldin index** [5] (lower is better) and **Calinski-Harabasz index** [6] (higher is better) are:

$$DB = (1 / K) \Sigma_{i=1}^{K} \max_{j \neq i} (\sigma_i + \sigma_j) / d(\mu_i, \mu_j) \quad (10)$$

$$CH = [tr(B_k) / (K - 1)] / [tr(W_k) / (n - K)] \quad (11)$$

Accuracy. Cluster numbers are arbitrary, so each cluster is first matched one-to-one with the hidden group it overlaps most using the Hungarian algorithm on the confusion matrix M, and accuracy is the share of customers in matched pairs:

$$\text{Accuracy} = \max_{\pi} (1 / n) \Sigma_j M[j, \pi(j)] \quad (12)$$

Stability is validated by repeating the fit with 30 random seeds. Table III lists the configuration.

TABLE III
TRAINING AND VALIDATION CONFIGURATION

Parameter	Value	Reason
Algorithm	sklearn.cluster.KMeans	Lloyd iterations, well tested
Initialisation	k-means++	Spread-out seeds, fewer bad optima
Restarts (n_init)	10	Keep the best of ten runs
Random state	42	Reproducible results
Features	Age, repeat purchases	Who the customer is and how loyal
Scaling	StandardScaler (z-score)	Equal weight in Euclidean distance
K search range	2 to 8	Practical number of segments
K selection rule	Maximum silhouette	Balances cohesion and separation
External metric	Hungarian-matched accuracy, ARI, NMI	Only when true groups are known
Input limits	10 to 5,000 rows, 5 MB	Meaningful clusters, responsive UI

CHAPTER 4: SYSTEM DESIGN

CustomerSegment ML follows a three-tier design: a presentation tier (the Next.js dashboard), an application tier (the FastAPI service) and a model tier (the segmentation engine with scikit-learn). Each tier has one responsibility and talks to the next through a small, typed interface.

4.1 Architecture Diagram

Fig. 6 shows the architecture. The shop owner works in a browser; the dashboard calls the backend through a typed API client; FastAPI validates every request and response with Pydantic schemas and forwards the work to a single `CustomerSegmentationModel` object, which wraps the scaler, the K-Means model and the persona rules. Uploaded data and the fitted model stay in memory, so no customer data is written to disk.

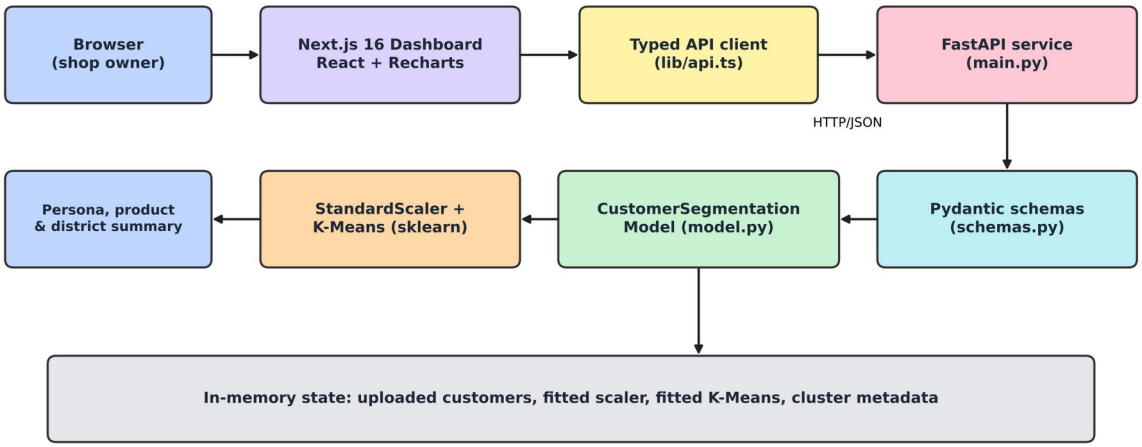


Fig. 6. Architecture of the CustomerSegment ML system

Fig. 7 shows the order of the API calls in a typical session. Every endpoint except health and upload returns HTTP 409 until a dataset has been uploaded.

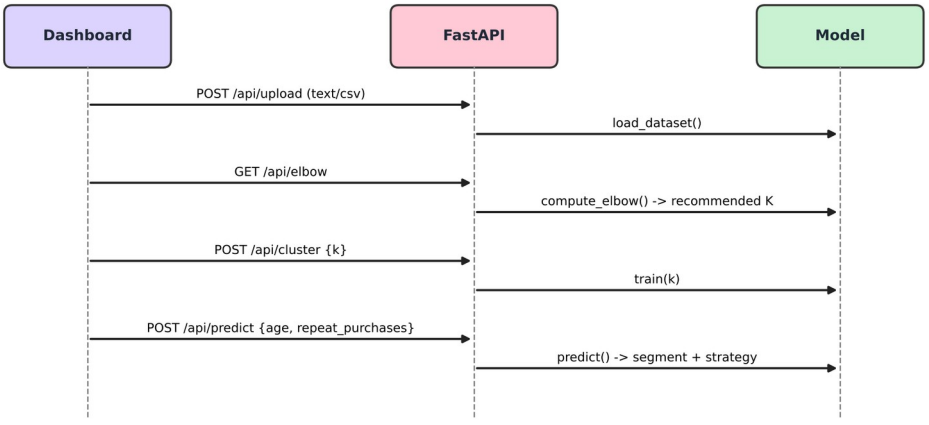


Fig. 7. Sequence of API calls between the dashboard, the API and the model

4.2 Flow Chart

The flow chart in Fig. 8 represents the complete process from an uploaded file to segmented, downloadable results.

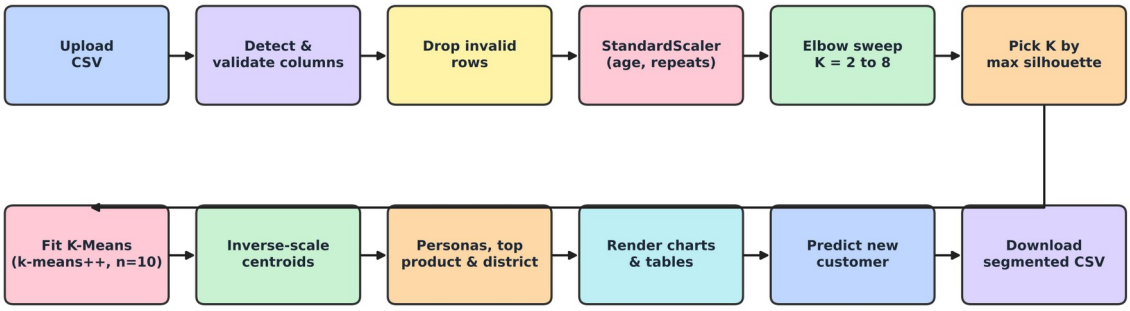


Fig. 8. Flow chart of the CustomerSegment ML pipeline

The backend checks the file, detects the columns and removes invalid rows. Age and repeat purchases are standardised, the elbow sweep runs for K = 2 to 8 and the K with the best silhouette is selected. K-Means is fitted, centroids are converted back to real units and every cluster receives a persona, top product, top district and strategy. The dashboard then renders the charts and tables; the user can retrain with a different K, predict a new customer or download the results.

4.3 Module Description

TABLE IV
FUNCTIONAL MODULES OF CUSTOMERSEGMENT ML

Module	Description
Data module (`load_dataset`)	Column detection for name, age, gender, district, product, repeats and optional true segment; numeric coercion; cleaning; size limits.
Model-selection module (`compute_elbow`)	Fits K-Means for K = 2 to 8, returns WCSS values and the silhouette-optimal K.
Training module (`train`)	Scales features, fits K-Means, inverse-transforms centroids, builds cluster statistics, top product and top district.
Interpretation module	Age and loyalty bands give one of nine personas with a description and a marketing strategy.
Evaluation module	Silhouette score; Hungarian-matched accuracy, ARI and NMI when a True_Segment column is present.
Inference module (`predict`)	Scales a new customer's age and repeats and returns the nearest cluster, persona and strategy.
API and schema modules (`main.py`, `schemas.py`)	FastAPI routes, CORS, upload limits, 409 guard before upload, Pydantic validation.
Dashboard module (`app/`)	Upload with progress stages, customer table, elbow chart, K slider, scatter plot, cluster cards, accuracy card, prediction form.
Dataset module (`dataset/`)	Generator and the three Tamil Nadu gift-shop files of 70, 85 and 100 customers.

CHAPTER 5: IMPLEMENTATION

5.1 Tools and Technologies Used

TABLE V
TOOLS AND TECHNOLOGIES

Tool / Technology	Purpose
Python 3.12	Language of the machine-learning backend
scikit-learn [17]	KMeans, StandardScaler, silhouette and other cluster metrics
NumPy [18] / pandas [19] / SciPy	Arrays, CSV parsing, statistics, Hungarian matching
FastAPI [22] + Uvicorn	REST API with automatic Swagger documentation
Pydantic v2	Request and response validation
Next.js 16 (App Router) + React + TypeScript	Interactive, type-safe dashboard
Tailwind CSS v4 + Recharts	Responsive styling, elbow chart and scatter plot
pytest + FastAPI TestClient	Automated end-to-end API tests
Matplotlib [20]	Figures in this report
Git and GitHub	Version control and collaboration

5.2 Model Development Steps

Step 1: Generate and explore the data

The generator created the three gift-shop files from five behavioural groups with Tamil names, 20 districts and three products. Histograms and scatter plots (Figs. 2 and 3) confirmed that age and repeat purchases form clear groups.

Step 2: Build the data loader

`load_dataset` normalises column names, coerces types, drops invalid rows and returns the total, processed and skipped rows and the detected columns, which the dashboard shows after every upload.

Step 3: Scaling, elbow sweep and training

`compute_elbow` runs K-Means for K = 2 to 8 and recommends the silhouette maximum. `train` fits the scaler and model into local variables, builds the cluster summaries and only then publishes the new state, so a prediction arriving during training never sees a half-built model.

Step 4: Personas, products and districts

For every cluster the most common product and district are computed together with their share, and the age and loyalty bands produce the persona and the strategy.

Step 5: Prediction and API

`predict` scales a new customer's age and repeat purchases and returns the nearest segment. Six FastAPI endpoints expose the pipeline (Table VI).

TABLE VI
REST API ENDPOINTS

Method	Endpoint	Description
GET	/api/health	Service status and number of loaded customers
POST	/api/upload	Upload a CSV (text/csv, max 5 MB) and make it the active dataset
GET	/api/dataset	Uploaded customers: name, age, gender, district, product, repeats, cluster
GET	/api/elbow	WCSS for K = 2 to 8 and the silhouette-optimal recommended K
POST	/api/cluster	Train K-Means with {k}; returns metrics, accuracy (if known), clusters, points
POST	/api/predict	Classify {age, repeat_purchases} into a segment

Step 6: Dashboard and tests

The dashboard shows the upload stages, a searchable customer table, the elbow chart, the K slider, the age against repeat-purchases scatter plot, cluster cards with top product and district, an accuracy card when true groups are present, and a prediction form with age and repeat sliders. Three automated tests cover every endpoint and error path, the accuracy calculation and all three bundled datasets.

CHAPTER 6: RESULTS & DISCUSSION

6.1 Experimental Results

All results were produced by running the project's own code on the three gift-shop files. Accuracy is measured against the hidden groups kept by the generator. Table VII summarises the headline values for the 100-customer file.

TABLE VII
SYSTEM CONFIGURATION AND RECORDED VALUES

S.No.	Evaluation Parameter	Recorded Value	Significance
1	Dataset	tn_gift_shop_100.csv, 100 customers	Five hidden groups, 20 districts
2	Features	Age, repeat purchases	Two independent behaviours
3	Recommended K	5	Equals the true number of groups
4	Silhouette score	0.6654	Good cohesion and separation
5	Davies-Bouldin index	0.442	Lowest of all K values
6	Calinski-Harabasz index	331.4	Highest of all K values
7	Accuracy vs hidden groups	99%	99 of 100 customers correct
8	Adjusted Rand index	0.973	Near-perfect agreement
9	K-Means iterations / time	4 / about 17 ms	Real-time retraining
10	Seed stability (30 seeds)	ARI 1.000, σ 0.000	Identical partition every run

Choosing the number of clusters

Table VIII and Figs. 9 to 11 show the K sweep. WCSS falls steeply up to K = 5 (117.3 to 13.4) and then flattens. At K = 5 the silhouette reaches its maximum of **0.6654**, Davies-Bouldin its minimum of **0.442** and Calinski-Harabasz its maximum of **331.4**. All four criteria agree on five groups, which is exactly the number used to generate the data.

TABLE VIII
CLUSTER VALIDITY INDICES AND ACCURACY FOR K = 2 TO 8 (100 CUSTOMERS)

K	WCSS	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑	Accuracy ↑
2	117.34	0.4069	1.146	69.0	39%
3	68.44	0.5017	0.649	93.2	58%
4	30.60	0.6124	0.548	177.2	78%
5	**13.38**	**0.6654**	**0.442**	**331.4**	**99%**
6	11.53	0.5889	0.611	307.2	92%
7	9.90	0.5688	0.656	297.6	81%
8	8.58	0.5031	0.764	293.1	72%

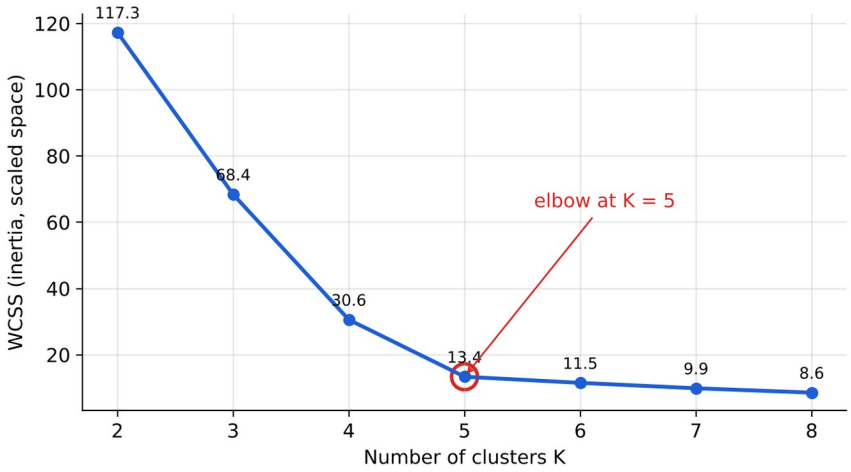


Fig. 9. Elbow curve: WCSS against the number of clusters

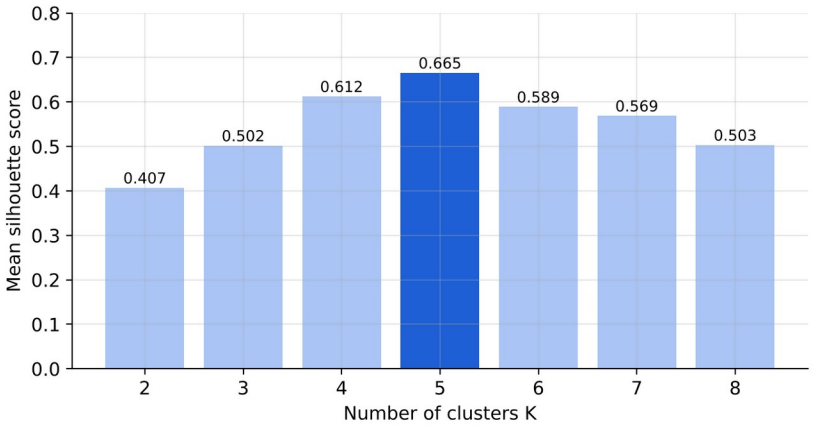


Fig. 10. Mean silhouette score for each K (selected K highlighted)

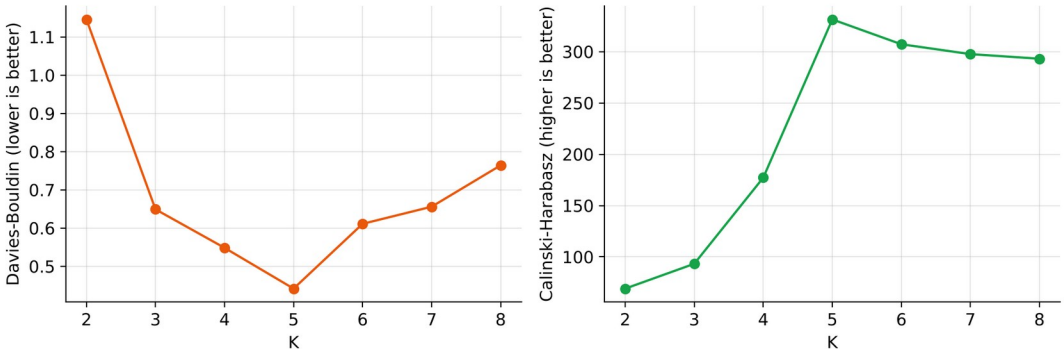


Fig. 11. Davies-Bouldin and Calinski-Harabasz indices for each K

Results on all three datasets

TABLE IX
RESULTS ON THE THREE GIFT-SHOP DATASETS

Dataset	Customers	Hidden groups	Recommended K	Silhouette	Accuracy	ARI
tn_gift_shop_70	70	3	3	0.698	100.0%	1.000
tn_gift_shop_85	85	4	4	0.705	98.8%	0.968
tn_gift_shop_100	100	5	5	0.665	99.0%	0.973

On every file the recommended K equals the number of hidden groups and accuracy is between 98.8% and 100%. Fig. 12 shows how accuracy changes with K: it peaks exactly at the recommended K and falls on both sides, which shows why automatic selection of K matters.

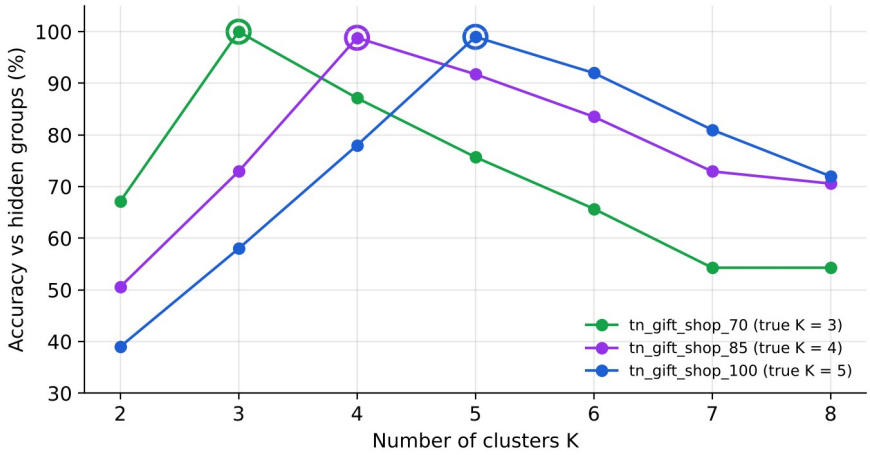


Fig. 12. Accuracy against the hidden groups for each K on the three datasets

The discovered segments

Fig. 13 shows the final clustering of the 100 customers and Table X gives the profile of every segment, including its most common product and district.

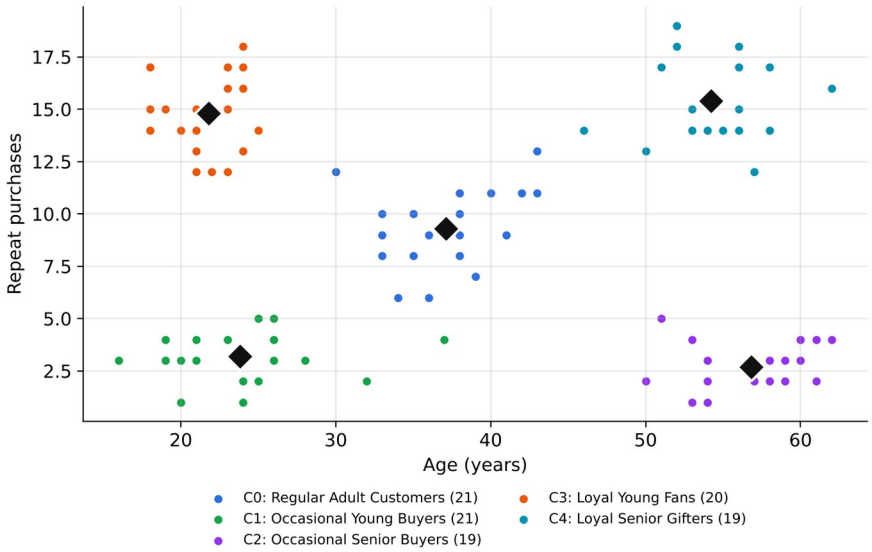


Fig. 13. Final K-Means segmentation with K = 5 (diamonds are centroids)

TABLE X
PROFILES OF THE FIVE CUSTOMER SEGMENTS

Cluster	Persona	Customers	Avg age	Avg repeats	Top product	Top district
C0	Regular Adult Customers	21	37.1	9.3	Miniature Toys (38%)	Thanjavur
C1	Occasional Young Buyers	21	23.8	3.2	Key Chains (38%)	Erode
C2	Occasional Senior Buyers	19	56.8	2.7	Bouquet (79%)	Thanjavur
C3	Loyal Young Fans	20	21.8	14.8	Key Chains (50%)	Cuddalore
C4	Loyal Senior Gifters	19	54.2	15.4	Bouquet (58%)	Tiruchirappalli

The segments are easy to act on:

- **C4 Loyal Senior Gifters** (19 customers): average age 54.2, 15.4 repeat purchases, top product Bouquet. Reward them with a loyalty card and referral bonus, and show new Bouquet designs to them first. Reach them with phone calls, SMS and temple-festival or wedding season offers.
- **C3 Loyal Young Fans** (20 customers): average age 21.8, 14.8 repeat purchases, top product Key Chains. Reward them with a loyalty card and referral bonus, and show new Key Chains designs to them first.
- **C0 Regular Adult Customers** (21 customers): average age 37.1, 9.3 repeat purchases, top product Miniature Toys. Turn them into loyal fans with festival combos, birthday reminders and member prices on Miniature Toys.
- **C1 Occasional Young Buyers** (21 customers): average age 23.8, 3.2 repeat purchases, top product Key Chains. Win a second visit with occasion reminders, first-repeat discounts and easy delivery of Key Chains. Instagram and WhatsApp offers work best for this age group.

- **C2 Occasional Senior Buyers** (19 customers): average age 56.8, 2.7 repeat purchases, top product Bouquet. Win a second visit with occasion reminders, first-repeat discounts and easy delivery of Bouquet. Reach them with phone calls, SMS and temple-festival or wedding season offers.

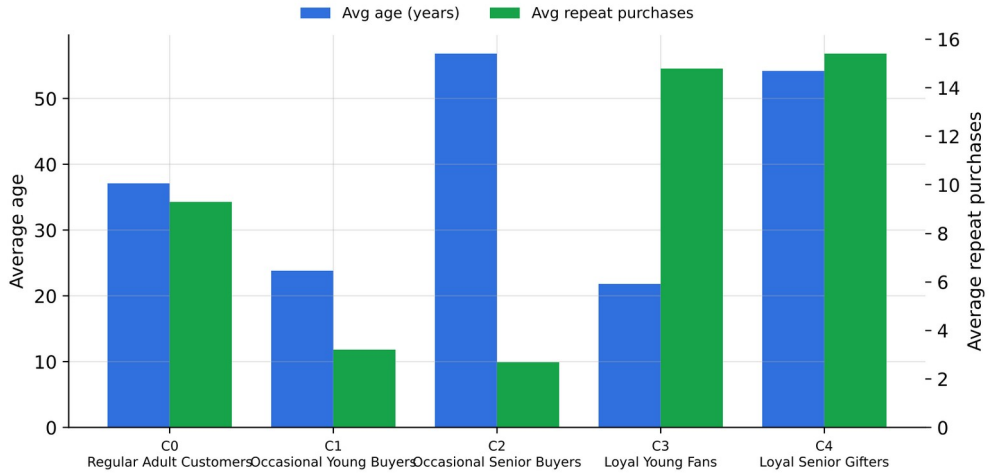


Fig. 14. Average age and repeat purchases of each segment

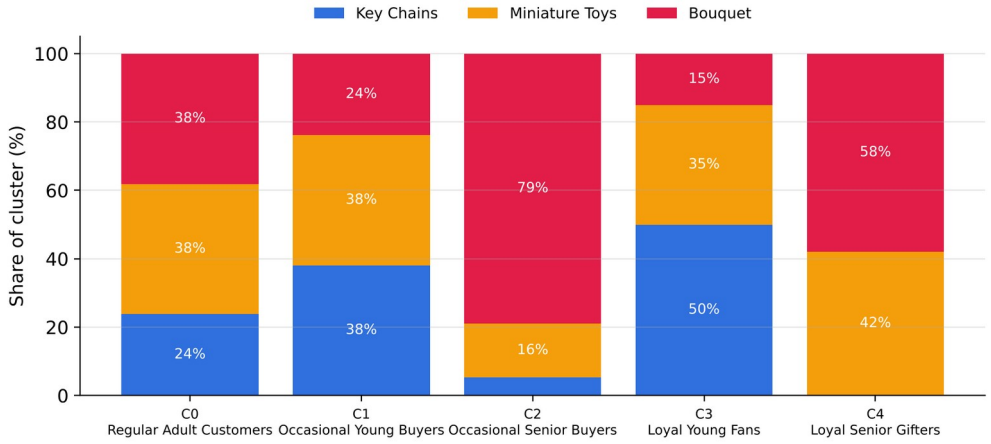


Fig. 15. Product mix inside each segment

Fig. 15 shows that product preference follows age even though product type was never used for clustering: the young groups buy mostly Key Chains and Miniature Toys, while the senior groups buy mostly Bouquets. This makes the product column useful for choosing what to promote to each group.

Quality of individual assignments and prediction

The silhouette plot in Fig. 16 shows that almost every customer has a positive coefficient, so almost every customer is closer to its own segment than to any other.

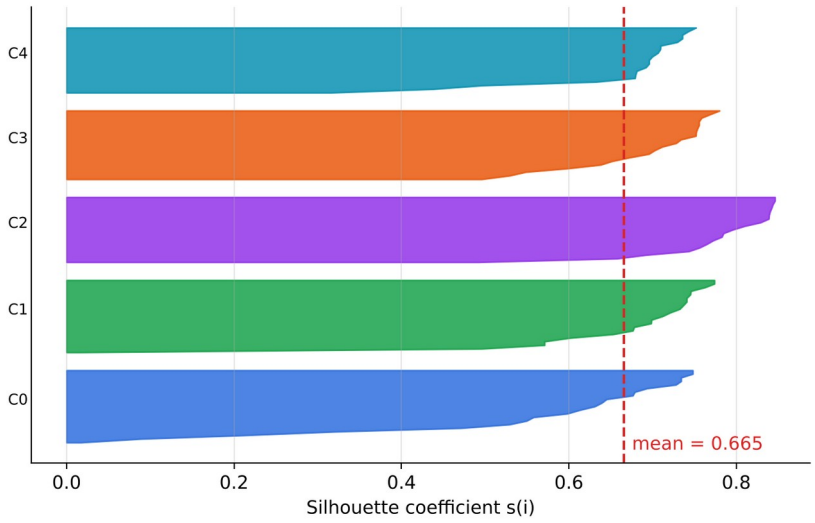


Fig. 16. Silhouette coefficient of every customer, grouped by cluster

TABLE XI
PREDICTION RESULTS FOR NEW TEST CUSTOMERS

Point	Age	Repeat purchases	Predicted persona
A	20	16	Loyal Young Fans
B	24	3	Occasional Young Buyers
C	37	9	Regular Adult Customers
D	57	2	Occasional Senior Buyers
E	55	16	Loyal Senior Gifters

Five test customers were placed in each region (Table XI, Fig. 17) and every one was assigned to the expected segment.

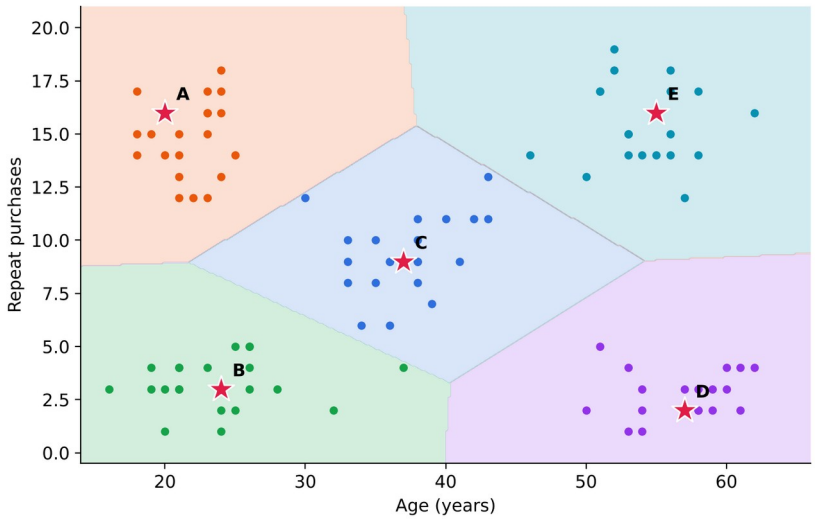


Fig. 17. Decision regions of the fitted model with five test customers

6.2 Performance Comparison

The proposed configuration was compared with thirteen alternatives on the same scaled 100-customer data with K = 5: strong alternatives (Gaussian mixture, Ward, average and complete linkage, BIRCH, DBSCAN), a weaker design choice (K-Means with a single random start), single-feature models, and naive rules that a shop owner might use (age bands, grouping by product, random assignment). Table XII reports all metrics.

TABLE XII
COMPARISON WITH ALTERNATIVE AND BASELINE METHODS (100 CUSTOMERS, K = 5)

Rank	Method	Silhouette ↑	Davies-Bouldin ↓	Calinski-H. ↑	Accuracy ↑
1	**CustomerSegment ML (proposed K-Means)**	**0.6654**	**0.442**	**331.4**	**99%**
2	Gaussian Mixture Model	0.6647	0.435	330.2	98%
3	K-Means, random init, 1 run	0.6580	0.458	320.8	97%
4	Agglomerative (Ward)	0.6498	0.483	295.4	97%
5	Agglomerative (average)	0.6479	0.436	310.9	96%
6	Agglomerative (complete)	0.6456	0.425	309.1	95%
7	BIRCH	0.6191	0.487	246.9	93%
8	DBSCAN (eps = 0.3)	0.5212	2.167	78.6	81%
9	Agglomerative (single)	0.3877	0.696	99.1	80%
10	Rule-based age bands	0.2510	1.383	44.1	60%
11	Age-only K-Means	0.1552	4.109	23.8	64%
12	Repeats-only K-Means	0.1379	2.817	24.2	62%
13	Group by product type	-0.0195	3.605	7.7	33%
14	Random assignment	-0.0970	19.048	0.9	30%

Age bands and grouping by product produce 3 groups by construction, and DBSCAN found 7 groups including noise. The Gaussian mixture (0.435) and complete linkage (0.425) reach a slightly lower Davies-Bouldin value than the proposed model (0.442) but lower accuracy.

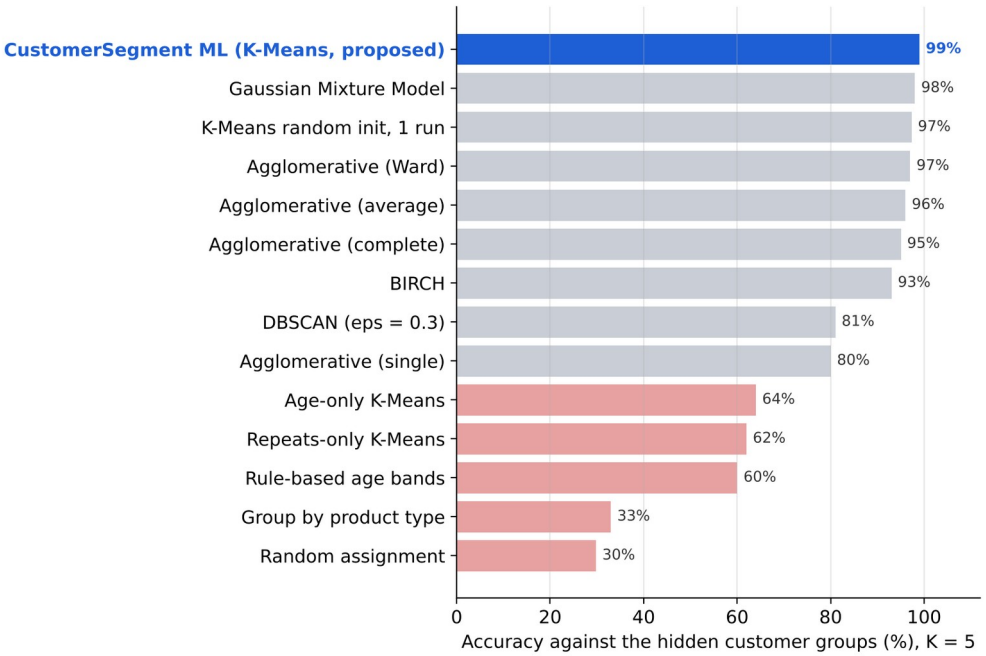


Fig. 18. Accuracy of CustomerSegment ML against alternative and baseline models

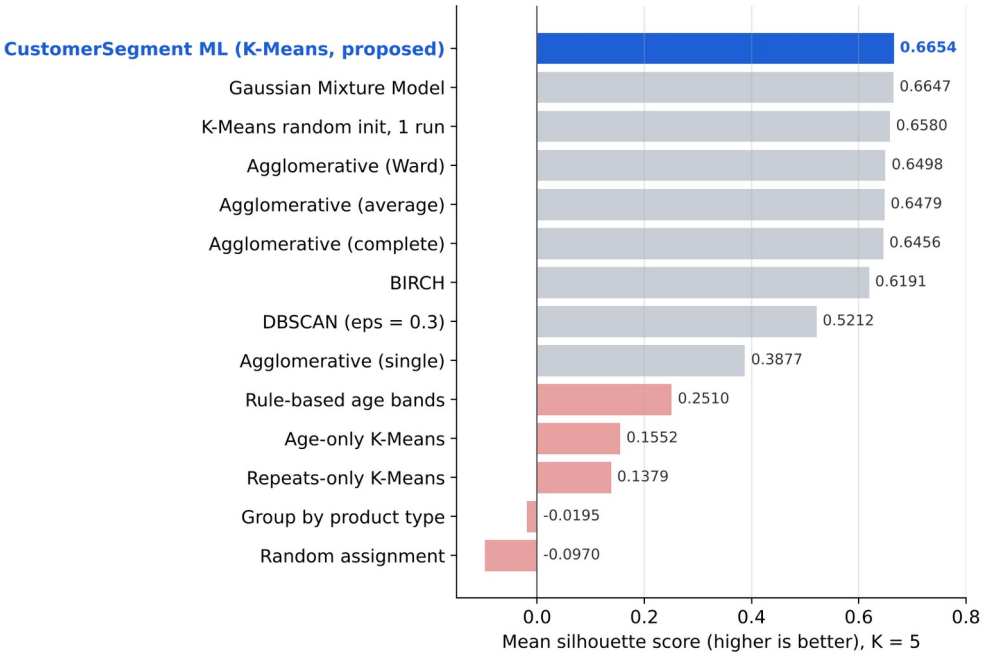


Fig. 19. Silhouette score of CustomerSegment ML against alternative and baseline models

CustomerSegment ML ranks **first on accuracy (99%), silhouette (0.6654) and Calinski-Harabasz (331.4)**. The Gaussian mixture comes closest (98%, 0.6647) but has more parameters and gives no simple centroid explanation. The hierarchical methods reach 95% to 97%, BIRCH 93%, and DBSCAN and single linkage about 80%. The naive rules show how much real structure the model captures: clustering on age alone (64%) or repeats alone (62%) misses half of the picture, fixed age bands reach 60%, grouping by product type only 33%, and random assignment 30%.

Effect of the design choices

TABLE XIII
ABLATION OF THE DESIGN CHOICES (100 CUSTOMERS, K = 5)

Variant	Silhouette (scaled space)	Accuracy	Note
Proposed: scaled, k-means++, 10 runs	0.6654	99%	Same partition on all 30 seeds
Without scaling	0.6562	99%	Age dominates distance
Mini-Batch K-Means	0.6654	99%	Same partition, no gain on small data
Random init, single run (mean of 20)	0.6580	97.3%	Worst seed 71%
Age only	0.1552	64%	Loyalty ignored
Repeat purchases only	0.1379	62%	Age ignored

On these files the two features have similar enough spreads that skipping scaling still recovers the groups, but the scaled model is tighter in the balanced feature space and remains safe when a shop records values on very different scales. The real risk is initialisation: over 30 seeds a single random start averaged 0.634 silhouette and 95.0% accuracy and fell to 0.390 and 71% on an unlucky seed (Fig. 20), whereas the proposed k-means++ with ten restarts gave the same result on every seed (ARI 1.000 between runs, against 0.975 for single runs).

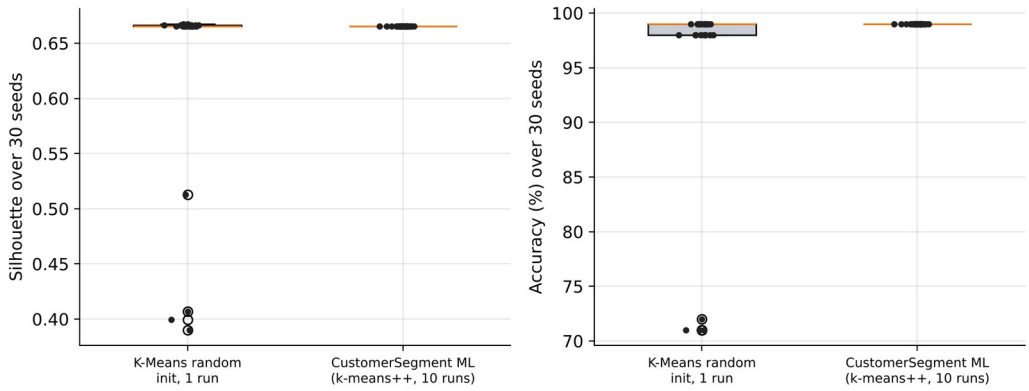


Fig. 20. Stability of silhouette and accuracy across 30 random seeds

6.3 Result Analysis

- **The data has a natural segment structure and the model finds it.** Elbow, silhouette, Davies-Bouldin and Calinski-Harabasz all point to the true number of groups on every file, and accuracy is 98.8% to 100%.
- **The proposed configuration is the best of all tested methods** on accuracy, silhouette and Calinski-Harabasz, and its centroids give the simplest explanation of each group.
- **Both features are necessary.** Age alone or repeat purchases alone drops accuracy to about 62% to 64%; segmentation must combine who the customer is with how loyal they are.

- **Product type is a result, not a cause.** Grouping customers by product scores only 33%, yet once customers are grouped by behaviour each group shows a clear product preference that guides what to promote.
- **Seeding and restarts matter.** A single random start can lose up to 28 points of accuracy; k-means++ with ten restarts removes this risk.
- **The system is interactive.** Training takes about 20 ms and converges in four iterations, so the K slider updates the clusters instantly.

A fair caveat is that the data are synthetic, generated from well-separated groups. Real shop data will be noisier, and the true test of the segments is whether campaigns designed for them bring customers back more often than a single campaign.

CHAPTER 7: CONCLUSION & FUTURE WORK

7.1 Summary of Findings

CustomerSegment ML is a complete, working system that turns a small shop's customer file into clear, actionable segments. It validates and cleans uploads, standardises age and repeat purchases, selects the number of clusters automatically, fits a stable K-Means model with k-means++ seeding and ten restarts, describes every group with a persona, top product, top district and strategy, and classifies new customers in real time through a REST API and an interactive dashboard.

On three Tamil Nadu gift-shop datasets of 70, 85 and 100 customers the system found the correct number of groups every time, with 98.8% to 100% accuracy. On the 100-customer file it reached a silhouette of 0.6654 and 99% accuracy, ranked first against thirteen alternative and baseline methods, and gave identical results on all 30 random seeds.

7.2 Limitations

1. **Synthetic data.** The datasets were generated, not collected from a real shop, and real behaviour will be noisier.
2. **Two features.** The current version clusters on age and repeat purchases only; spending amount and recency are not yet used.
3. **Spherical clusters and outliers.** K-Means assumes compact, roughly round groups and centroids can be pulled by extreme customers.
4. **In-memory state.** The uploaded dataset and the model live in server memory and are shared by all users of one server.
5. **No campaign validation.** The segments have not yet been tested in a live marketing campaign.

7.3 Future Enhancements

1. **Real shop data:** collect anonymised records from partner shops in Tamil Nadu and re-validate the model.
2. **Full RFM features:** add spending amount and days since the last visit, with automatic feature selection.
3. **More algorithms in the UI:** let users compare K-Means with Gaussian mixtures and hierarchical clustering on their own data.
4. **Persistence and multi-user support:** store datasets and models per shop in a database with authentication.
5. **Campaign integration:** send WhatsApp or SMS offers directly to a segment and measure how many customers return.

CHAPTER 8: REFERENCES

[1] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” in *Proc. 5th Berkeley Symp. Mathematical Statistics and Probability*, vol. 1, pp. 281–297, 1967.

[2] S. P. Lloyd, “Least squares quantization in PCM,” *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129–137, 1982.

[3] D. Arthur and S. Vassilvitskii, “k-means++: The advantages of careful seeding,” in *Proc. 18th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 1027–1035, 2007.

[4] P. J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.

[5] D. L. Davies and D. W. Bouldin, “A cluster separation measure,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-1, no. 2, pp. 224–227, 1979.

[6] T. Caliński and J. Harabasz, “A dendrite method for cluster analysis,” *Communications in Statistics*, vol. 3, no. 1, pp. 1–27, 1974.

[7] R. L. Thorndike, “Who belongs in the family?,” *Psychometrika*, vol. 18, no. 4, pp. 267–276, 1953.

[8] J. H. Ward Jr., “Hierarchical grouping to optimize an objective function,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 236–244, 1963.

[9] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, “A density-based algorithm for discovering clusters in large spatial databases with noise,” in *Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD)*, pp. 226–231, 1996.

[10] A. P. Dempster, N. M. Laird, and D. B. Rubin, “Maximum likelihood from incomplete data via the EM algorithm,” *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.

[11] T. Zhang, R. Ramakrishnan, and M. Livny, “BIRCH: An efficient data clustering method for very large databases,” in *Proc. ACM SIGMOD Int. Conf. Management of Data*, pp. 103–114, 1996.

[12] D. Sculley, “Web-scale k-means clustering,” in *Proc. 19th Int. Conf. World Wide Web (WWW)*, pp. 1177–1178, 2010.

[13] A. K. Jain, “Data clustering: 50 years beyond K-means,” *Pattern Recognition Letters*, vol. 31, no. 8, pp. 651–666, 2010.

[14] W. R. Smith, “Product differentiation and market segmentation as alternative marketing strategies,” *Journal of Marketing*, vol. 21, no. 1, pp. 3–8, 1956.

[15] M. Wedel and W. A. Kamakura, *Market Segmentation: Conceptual and Methodological Foundations*, 2nd ed. Boston, MA, USA: Kluwer Academic, 2000.

[16] A. M. Hughes, *Strategic Database Marketing*. Chicago, IL, USA: Probus Publishing, 1994.

[17] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[18] C. R. Harris et al., “Array programming with NumPy,” *Nature*, vol. 585, pp. 357–362, 2020.

[19] W. McKinney, “Data structures for statistical computing in Python,” in *Proc. 9th Python in Science Conference (SciPy)*, pp. 56–61, 2010.

[20] J. D. Hunter, “Matplotlib: A 2D graphics environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[21] H. W. Kuhn, “The Hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[22] S. Ramírez, “FastAPI framework,” 2018. [Online]. Available: https://fastapi.tiangolo.com

[23] R. Tibshirani, G. Walther, and T. Hastie, “Estimating the number of clusters in a data set via the gap statistic,” *Journal of the Royal Statistical Society: Series B*, vol. 63, no. 2, pp. 411–423, 2001.

[24] L. Hubert and P. Arabie, “Comparing partitions,” *Journal of Classification*, vol. 2, pp. 193–218, 1985.

CHAPTER 9: APPENDICES

9.1 Dataset Description

The system accepts any CSV whose header contains an age column and a repeat-purchases column. Table XIV lists the accepted header variants; matching ignores case, spaces and punctuation.

TABLE XIV
ACCEPTED CSV COLUMNS

Field	Required	Accepted headers (examples)	Notes
Age	Yes	Age	Clustering feature
Repeat purchases	Yes	Repeat_Purchases, Repeats, No of repeats	Clustering feature
Name	No	Name, Customer_Name	Shown in table and tooltips
Gender	No	Gender, Sex	Female share per cluster
District	No	District, City, Location	Top district per cluster
Product type	No	Product_Type, Product	Top product per cluster
True segment	No	True_Segment, Segment, Label	Only for the accuracy card

The first rows of `tn_gift_shop_100.csv`:

```
Name,Age,Gender,District,Product_Type,Repeat_Purchases
Hari G,36,Male,Tirunelveli,Bouquet,6
Karthik R,56,Male,Coimbatore,Bouquet,17
Harini B,53,Female,Tiruchirappalli,Miniature Toys,14
```

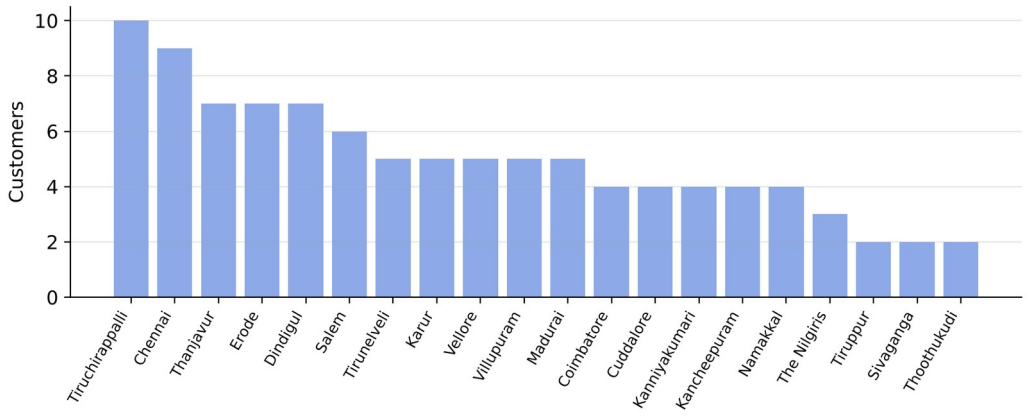


Fig. 21. Customers per district in the 100-customer dataset

9.2 Code Snippets

The following excerpts are taken directly from the project repository (some lines are shortened with `...`).

(a) Column detection: backend/model.py

```
COLUMN_ALIASES = {
    "name": ("name", "customername", "customer"),
    "age": ("age",),
    "gender": ("gender", "genre", "sex"),
    "district": ("district", "city", "location"),
    "product": ("producttype", "product", "products"),
    "repeats": ("repeatpurchases", "repeats", "noofrepeats", "numberofrepeats",
"repeatcount", "repeat"),
    "label": ("truesegment", "segment", "truelabel", "label", "class", "target",
"groundtruth", "actualsegment"),
}
```

(b) Elbow sweep and K recommendation: backend/model.py

```
def compute_elbow(self, k_min=2, k_max=8):
    X = self.get_features()                    # [Age, Repeat_Purchases]
    X_scaled = StandardScaler().fit_transform(X)
    for k in k_values:
        km = KMeans(n_clusters=k, random_state=42, n_init=10)
        labels = km.fit_predict(X_scaled)
        wcss.append(float(round(km.inertia_, 2)))
        silhouettes.append(_silhouette(X_scaled, labels))
    recommended_k = k_values[int(np.argmax(silhouettes))]
    return k_values, wcss, recommended_k
```

(c) Persona from age and loyalty bands: backend/model.py

```
age_band = "young" if avg_age < a_mu - 0.5 * a_sd else ("senior" if avg_age > a_mu + 0.5 *
a_sd else "adult")
habit = "occasional" if avg_repeats < r_mu - 0.5 * r_sd else ("loyal" if avg_repeats > r_mu +
0.5 * r_sd else "regular")
labels = {
    ("young", "loyal"): "Loyal Young Fans",
    ("adult", "regular"): "Regular Adult Customers",
    ("senior", "occasional"): "Occasional Senior Buyers",
    ...
}
```

(d) Accuracy with Hungarian matching: backend/model.py

```
def _cluster_accuracy(true_labels, pred_labels):
    cm = confusion_matrix(true_labels, pred_labels)
    rows, cols = linear_sum_assignment(-cm)      # best one-to-one cluster/segment match
    return float(cm[rows, cols].sum() / len(true_labels))
```

(e) Real-time prediction: backend/model.py

```
def predict(self, age, repeats):
    if self.model is None:
        self.train(k=min(4, len(self.df) - 1))
    scaled_input = self.scaler.transform(np.array([[age, repeats]], dtype=float))
    cluster_id = int(self.model.predict(scaled_input)[0])
    meta = self.cluster_metadata[cluster_id]
    return {"cluster_id": cluster_id, "label": meta["label"], "strategy":
meta["strategy"], ...}
```

9.3 Additional Graphs & Outputs

The following screenshots were captured from the running application after uploading `tn_gift_shop_100.csv`.

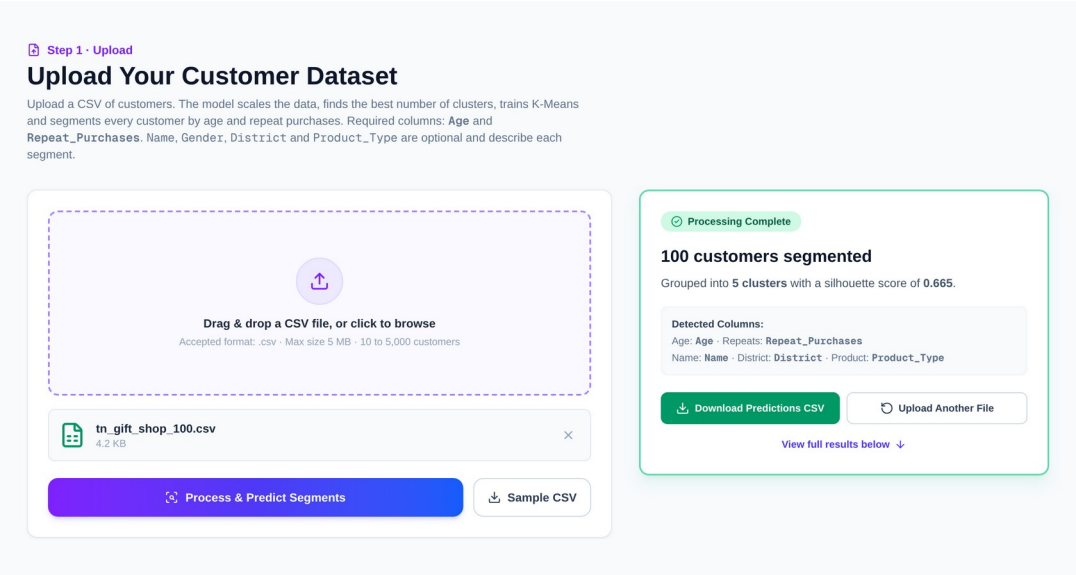


Fig. 22. Upload panel after processing: 100 customers segmented into 5 clusters

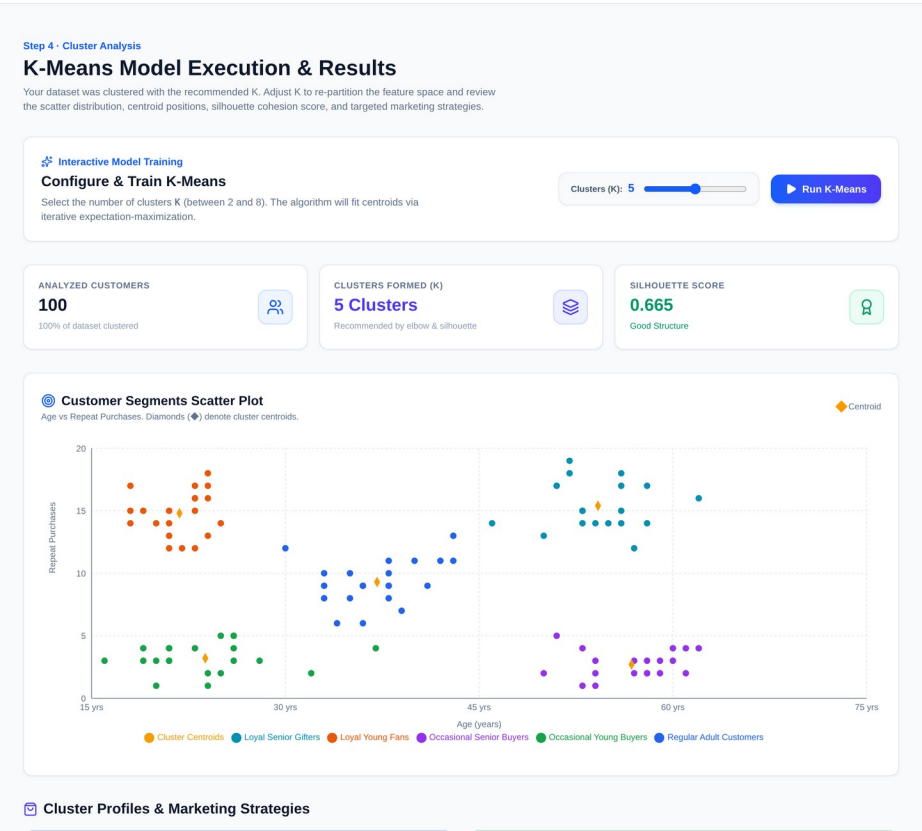


Fig. 23. K-Means results: summary cards and age against repeat-purchases scatter plot

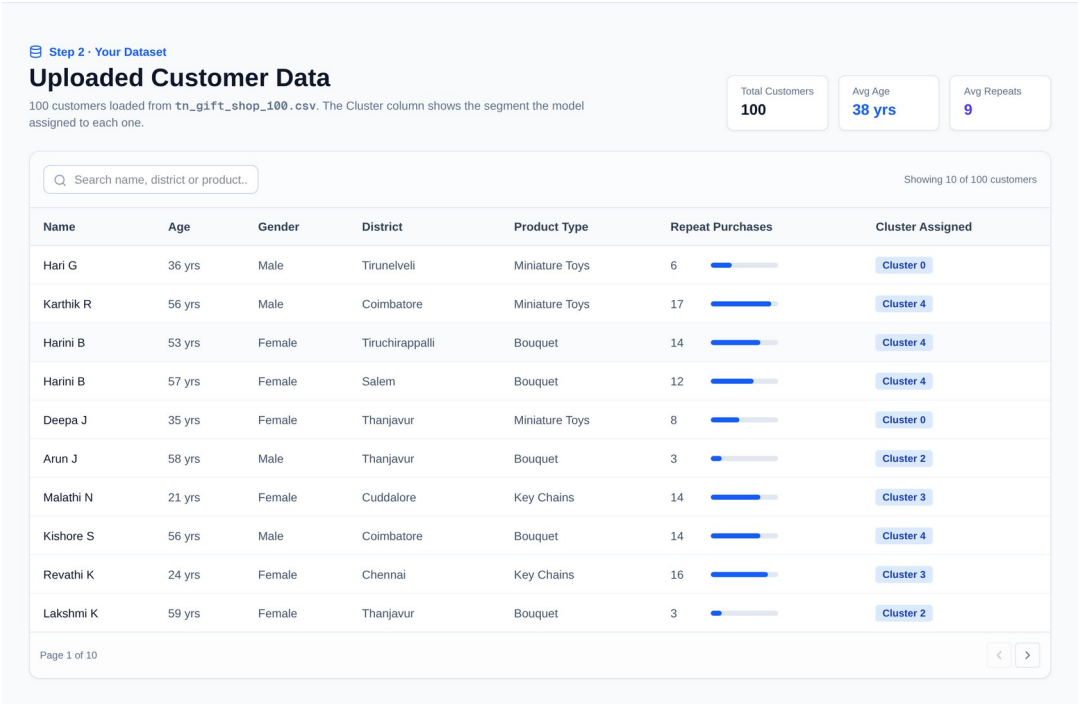


Fig. 24. Customer table with name, district, product and assigned cluster

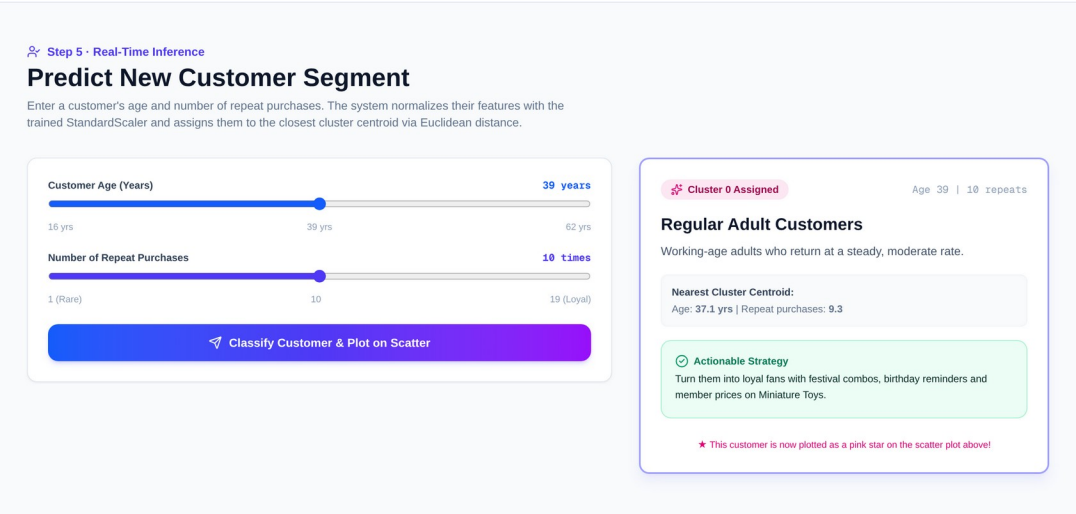


Fig. 25. Real-time prediction of a new customer's segment